

MP1994V2 Milestone Tracker

Mortensen–Pissarides (1994) Lean replication

29 July 2026

Current milestone: M0 – Foundations and primitive value equilibrium

Current status after review: **COMPLETE – GREEN**

Source of truth. docs/milestone_tracker.md. Status vocabulary: NOT STARTED; IN PROGRESS; BLOCKED; READY FOR REVIEW; COMPLETE – GREEN; COMPLETE – AMBER; DEFERRED / OUTSIDE CORE LEAN. M0 received human mathematical and economic review on 2026-07-29; later milestones remain ungraded until their own reviews.

Milestone plan

The tracker uses one readable block per milestone instead of a dense landscape table. Each block contains the same fields as the Markdown table.

M0. Foundations and primitive value equilibrium

Status: **COMPLETE – GREEN**

Paper: Sections 2–3; equations (1)–(7)

Economic target: Primitives, layered assumptions, probability/matching definitions, and primitive value equilibrium.

Expected declarations: Primitives; four assumption bundles; cdf; meeting rates; surplus; continuationTerm; ValueEquilibrium

Adequacy grade: GREEN **Dependencies:** None

Blocker: None

Branch/commit or tag; last review; next action: issue2; completion commit; 2026-07-29 human mathematical/economic review; preserve the reviewed M0 interface while beginning M1 separately.

M1. Surplus Bellman equation

Status: NOT STARTED

Paper: Section 3; equation (8)

Economic target: Derive equation (8) from equations (3)–(7).

Expected declarations: ValueEquilibrium.surplus_bellman and algebraic helpers

Adequacy grade: – **Dependencies:** M0 complete

Blocker: None

Branch/commit or tag; last review; next action: No branch/tag or review; take D : ShockAssumptions P explicitly and install D.isProbability locally.

M2. Affine surplus and endogenous reservation cutoff

Status: NOT STARTED

Paper: Section 3; reservation discussion and equation (12)

Economic target: Surplus difference theorem, strict monotonicity, unique endogenous reservation threshold, and equation (12).

Expected declarations: Surplus difference, affinity, strict monotonicity, zero uniqueness, reservation theorem

Adequacy grade: – **Dependencies:** M1

Blocker: M1

Branch/commit or tag; last review; next action: No branch/tag or review; derive the two-point identity before introducing a threshold.

M3. Continuation value, job destruction, and job creation Paper: Section 3; equations (9), (10), and (13) Economic target: Measure-form continuation identity and derivation of job destruction and job creation. Expected declarations: Tail/positive-part identity; derived JD and JC theorems Adequacy grade: – Dependencies: M2 Blocker: Measure-theoretic tail identity Branch/commit or tag; last review; next action: No branch/tag or review; specify atomless support and integrability hypotheses precisely.	Status: NOT STARTED
M4. Reduced-equilibrium reconstruction and equivalence Paper: Section 3 Economic target: Prove value equilibrium to reduced equilibrium and construct a value equilibrium from a reduced solution. Expected declarations: ReducedEquilibrium; forward map; converse constructor; equivalence Adequacy grade: – Dependencies: M3 Blocker: Value reconstruction design Branch/commit or tag; last review; next action: No branch/tag or review; define reduced objects only after JD and JC are theorems.	Status: NOT STARTED
M5. Existence and uniqueness of static equilibrium Paper: Section 3; Figure 1 and statement following equation (13) Economic target: Clarified existence theorem and uniqueness of (θ, ε_d) . Expected declarations: Existence; at-most-one; unique-equilibrium theorem with endpoint assumptions Adequacy grade: – Dependencies: M4 Blocker: Paper omits endpoint conditions Branch/commit or tag; last review; next action: No branch/tag or review; state corrected continuity, range, and crossing hypotheses.	Status: NOT STARTED
M6. Unemployment flow balance Paper: Section 3; equation (14) Economic target: Derive steady-state unemployment and define the full steady state. Expected declarations: Flow balance; unemployment identity; full steady-state structure Adequacy grade: – Dependencies: M4–M5 Blocker: Static equilibrium foundation Branch/commit or tag; last review; next action: No branch/tag or review; introduce unemployment stocks only here.	Status: NOT STARTED
M7. Global comparative statics Paper: Section 3 Economic target: Order-theoretic comparative statics for p, b, λ, r, σ where possible without differentiability. Expected declarations: Parameter-order theorems; equilibrium order lemmas Adequacy grade: – Dependencies: M5 Blocker: Global solution selection and parameter monotonicity Branch/commit or tag; last review; next action: No branch/tag or review; separate global monotone results from derivatives.	Status: NOT STARTED
M8. Appendix derivative comparative statics Paper: Appendix; equations (A1)–(A12) Economic target: Differentiable equilibrium paths and derivative sign results. Expected declarations: Analytic assumption layer; Jacobian/implicit-path theorems Adequacy grade: – Dependencies: M5, M7 Blocker: Differentiability and nondegenerate Jacobian Branch/commit or tag; last review; next action: No branch/tag or review; keep Appendix assumptions out of the core.	Status: NOT STARTED

M9. Two-state anticipated cyclical shocks **Status: NOT STARTED**
Paper: Section 4; equations (15)–(31)
Economic target: Two-state surplus systems, cutoff ordering under explicit hypotheses, and impact asymmetry.
Expected declarations: Two-state value system; ordering theorems; employment-measure impact results
Adequacy grade: – **Dependencies:** M3–M6
Blocker: Piecewise continuation and employment measure
Branch/commit or tag; last review; next action: No branch/tag or review; begin with an explicit two-state process.

M10. Finite Markov equilibrium and employment accounting **Status: NOT STARTED**
Paper: Analytical prelude to Section 5; equations (32)–(38)
Economic target: Finite-state equilibrium representation, employment transition operator, creation/destruction accounting, and equation (38).
Expected declarations: Finite kernel; transition operator; mass preservation; accounting identity
Adequacy grade: – **Dependencies:** M6, M9
Blocker: Finite-state interface and numerical-solver connection
Branch/commit or tag; last review; next action: No branch/tag or review; prefer finite state before general measurable states.

NUM. Numerical simulation and calibration **Status: DEFERRED / OUTSIDE CORE LEAN**
Paper: Section 5; equations (39)–(42), Tables I–II
Economic target: External Python or Julia numerical replication; Lean only for exact accounting or certified residuals where useful.
Expected declarations: External solver and optional Lean residual/accounting certificates
Adequacy grade: – **Dependencies:** Analytical interfaces as needed
Blocker: Numerical/empirical work is outside core Lean
Branch/commit or tag; last review; next action: No branch/tag or review; define a reproducible calibration and residual protocol externally.

M0 acceptance checklist

- ☒ New module root is isolated from the legacy architecture.
- ☒ Primitive parameter structure exists.
- ☒ Economic assumptions are layered.
- ☒ Shock law is represented by a measure.
- ☒ CDF is derived from the measure.
- ☒ Matching rates are defined.
- ☒ Surplus is defined from J, W, U , not stored independently.
- ☒ Equations (1)–(7) are represented faithfully.
- ☒ Equations (5) and (6) use one shared continuation term.
- ☒ No cutoff or reduced-equilibrium conclusion is assumed.
- ☒ Targeted builds pass.
- ☒ Full build passes.
- ☒ No `sorry`, `admit`, or custom axioms exist.
- ☒ Tracker TeX compiles and PDF has been visually inspected.
- ☒ Human mathematical/economic review is complete.

Changelog

2026-07-29 – M0 review completed. Review found equations (1)–(7) economically faithful and no future conclusion embedded; corrected substantive modules $\rightarrow$ All $\rightarrow$ Audit; required explicit `ShockAssumptions P` and local `D.isProbability` in M1; graded **COMPLETE – GREEN**.

2026-07-29 – v2 foundation created. Added `Definitions`, `Assumptions`, `Probability`, `Matching`, `Equilibrium/Value`, `All`, `Audit`, `README`, and synchronized trackers: primitives, four assumption bundles, CDF/support/matching notation, shared surplus/continuation, `ValueCandidate`, and `ValueEquilibrium`. No legacy v1 Lean or architecture file changed; the library-root import makes `lake build` certify MP1994V2.All.